

Notes: Moreira & Savov (JF, 2017)

The Macroeconomics of Shadow Banking

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1 Big picture

- **Question.** How does shadow banking expand liquidity in good times, why does it collapse when uncertainty rises, and how does this fragile liquidity transformation affect asset prices, investment, growth, and monetary policy?
- **Core idea.** Shadow banking creates securities that are money-like in normal times but become illiquid when uncertainty spikes.
- **Mechanism.** Intermediaries tranche risky assets into money, shadow money, and equity. Money is stable but collateral-intensive; shadow money economizes on collateral but is fragile.
- **Takeaway.** Shadow banking is fragile liquidity creation: a trade-off between abundant liquidity and financial stability.

2 Incremental contribution of the paper

- **Beyond net-worth models.** The key scarce object is collateral, not intermediary net worth. Equity can signal constraint because it replaces liquid funding when collateral is scarce.
- **Beyond bank-run models.** Fragility arises without multiple equilibria. Securities become information-sensitive when uncertainty rises.
- **Safe-asset/liquidity contribution.** The model endogenizes why money and shadow money differ in stability.
- **Policy contribution.** The paper explains why LSAP and Operation Twist can have opposite effects even if both operate through central-bank balance sheets.

3 Model primitives and measurement

Investors have liquidity events. Intermediaries buy assets and issue tranches. Liquid claims must be safe enough that they can be traded without adverse-selection discounts.

$$U_0 = \max E_0[z_1 C_1 + C_2], \quad m_0 + s_0(1 - \bar{\kappa}) \leq 1 - \kappa_Y.$$

Interpretation: The collateral constraint is the core measurement object. It says that liquid securities can only be issued against crash-state asset value.

3.1 Security types

- Money has zero crash exposure and is always liquid.
- Shadow money has intermediate crash exposure and is liquid only when uncertainty is low.
- Equity is the residual illiquid tranche and provides a cushion.

Interpretation: Shadow money expands liquidity by economizing on collateral, but this is exactly what makes the liquidity supply fragile.

4 Models and theoretical specifications

4.1 Collateral constraint and collateral premium

$$\begin{aligned} m_t + s_t(1 - \bar{\kappa}) &\leq 1 - K_{A,t} \quad [\theta_t], \\ 1 - K_{A,t} &= X_t(1 - K_t^a) + (1 - X_t)(1 - K_t^b). \end{aligned}$$

Interpretation: The multiplier θ_t is the collateral premium. It transmits collateral scarcity into funding costs and asset prices.

4.2 Boom-bust mechanism

Low uncertainty makes shadow money attractive. It expands liquidity, lowers discount rates, raises asset prices, and shifts investment toward risky capital. A rise in uncertainty reverses this mechanism: shadow money collapses, money demand rises, equity buffers expand, and liquidity contracts.

Interpretation: The boom increases growth but also reduces collateral quality, making the bust worse.

5 Calibration, numerical results, and policy design

The numerical model uses benchmark parameters in Table II. Figures 3–9 show how uncertainty, crashes, LSAP, and Operation Twist affect securities issuance, premia, collateral, prices, and output. These are theoretical counterfactuals, not causal empirical estimates.

6 Tables and figures

6.1 Figure 1: The Economy’s Balance Sheet; Table I: Summary of Variables

Read: Figure 1 maps real assets into intermediary liabilities: money, shadow money, and equity. Table I defines the model’s state variables, prices, premia, and returns.

Economic message: The model is about balance-sheet collateral and liquid claims, not only bank capital.

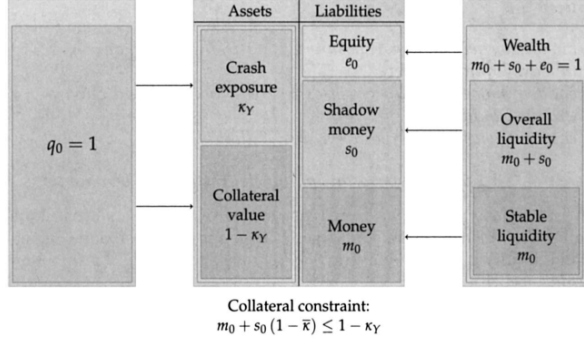


Figure 1. The economy's balance sheet. This figure represents the economy's balance sheet. The left side represents real assets (claims to output/capital), whose value is normalized to one in the static model. Intermediaries hold assets and issue money, shadow money, and equity against them, subject to the collateral constraint (6). Investors hold these securities. In the dynamic model, quantities receive a t subscript, the value of assets becomes the value of the aggregate capital stock, $q_t^a k_t^a + q_t^b k_t^b$, and collateral values become endogenous, $1 - \kappa_{A,t}$. (Color figure can be viewed at wileyonlinelibrary.com)

(a) Figure 1: The economy's balance sheet.

This table contains descriptions of the variables in the dynamic model in Section II.

Description	Variable
Investor:	
Consumption outside a liquidity event	dc_t
Consumption in a liquidity event	dC_t
Upper bound on liquidity event consumption	$\bar{dC}_t$
Wealth	W_t
Holdings of money, shadow money, and equity (% of wealth)	m_t, s_t, e_t
Technology:	
Compensated crash process	dZ_t
Uncertainty (perceived crash intensity)	λ_t
Capital stock	k_t^a, k_t^b
Investment rate	q_t^a, q_t^b
Risky capital share	χ_t
Output	Y_t
Asset markets:	
Price of assets	q_t^a, q_t^b
Drift and volatility of asset prices	$\mu_{q,t}^a, \mu_{q,t}^b, \sigma_{q,t}^a, \sigma_{q,t}^b$
Asset price crash exposure (% of price)	$\kappa_{q,t}^a, \kappa_{q,t}^b$
Asset collateral values (% of market value)	$1 - \kappa_{A,t}^a, 1 - \kappa_{A,t}^b$
Price-weighted risky capital share	χ_t^a
Aggregate collateral values (% of wealth)	$1 - \kappa_{A,t}$
Collateral premium	θ_t
Security markets:	
Returns on money, shadow money, and equity	dr_t^m, dr_t^s, dr_t^e
Expected return on money, shadow money, equity, and wealth	$\mu_{m,t}, \mu_{s,t}, \mu_{e,t}, \mu_{W,t}$

(b) Table I: Summary of variables.

6.2 Figure 2: Equilibrium Security Issuance; Table II: Benchmark Parameters

Read: Figure 2 shows the shift from shadow money to money as uncertainty rises. Table II lists the calibration used in the numerical exercises.

Economic message: Shadow money is profitable only when uncertainty is low enough that investors trust its liquidity.

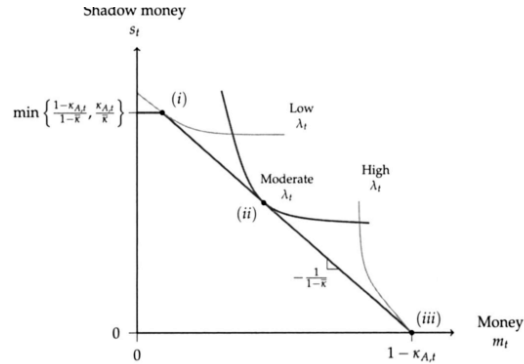


Figure 2. Equilibrium security issuance. This figure illustrates equilibrium issuance of shadow money s_t in the dynamic model as given in Proposition 3. The blue line is the liquidity provision frontier implied by the collateral constraint (22). The indifference curves at three different levels of uncertainty λ_t . Equilibrium is marked with black circles and labeled according to the three cases in Proposition 3. (Color figure can be viewed at wileyonlinelibrary.com)

(a) Figure 2: Equilibrium security issuance.

Benchmark Parameters

This table contains benchmark values for the model parameters used to produce results for the dynamic model. The investment cost function is parameterized as $\phi(i) = 1/\gamma(\sqrt{1+2\gamma i} - 1)$. We use the specification implied by the static model for the probability that shadow money becomes illiquid (i.e., $p_H(\lambda) = (\lambda - \lambda^L)/(\lambda^H - \lambda^L)$).

Description	Parameter	Value
Technology:		
Asset cash flows	y^a, y^b	0.138, 0.1
Depreciation rate	δ	0.1
Exogenous aggregate growth	μ_0	0.01
Adjustment cost parameter	γ	3
Asset crash exposures	κ_R^a, κ_R^b	0.5, 0
Information sensitivity constraint:		
Crash exposure limit for fragile liquid securities	$\bar{\kappa}$	0.7
Uncertainty:		
Low/high uncertainty states	λ^L, λ^H	0.005, 1
Average uncertainty	$\bar{\lambda}$	0.0245
Uncertainty rate of mean reversion	φ	0.5
Uncertainty news signal precision	$1/\sigma$	0.1
Preferences and liquidity events:		
Liquidity event frequency	h	0.28
Liquidity event marginal utility	ψ	5
Average size of liquidity event	$1/\eta$	0.33
Subjective discounting parameter	ρ	0.37

higher uncertainty causes demand for shadow money to fall and demand for

(b) Table II: Benchmark parameters.

6.3 Figure 3: Security Issuance and Expected Returns

Read: Shadow money is abundant at low uncertainty and collapses at high uncertainty; money and equity rise as stable liquidity requires more collateral and loss-absorbing buffers.

Economic message: The figure summarizes the endogenous shadow-banking cycle.

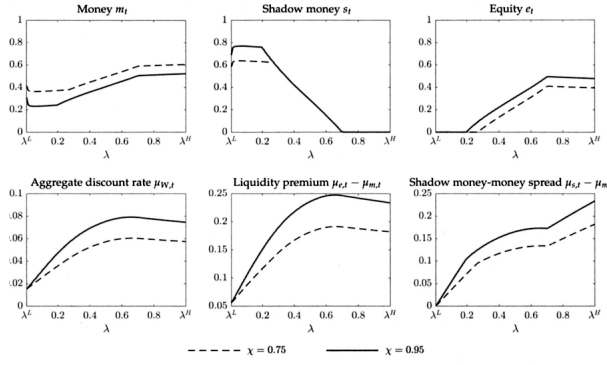


Figure 3: Security issuance and expected returns.

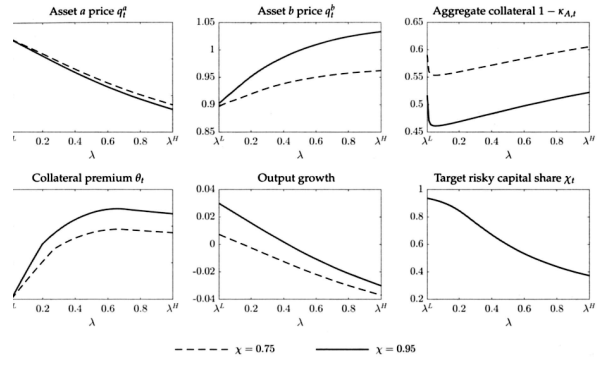


Figure 4: Asset prices, collateral, and economic activity.

6.4 Figure 4: Asset Prices, Collateral, and Economic Activity

Read: Risky asset prices and output growth fall with uncertainty, while safe assets appreciate because they become valuable collateral.

Economic message: Financial fragility spills into the real economy through discount rates, collateral values, and investment.

6.5 Figure 5: The Economy with and without Shadow Banking

Read: Shadow banking raises liquidity and asset prices in quiet times but makes the economy more fragile when uncertainty rises.

Economic message: This figure displays the paper's central welfare trade-off.

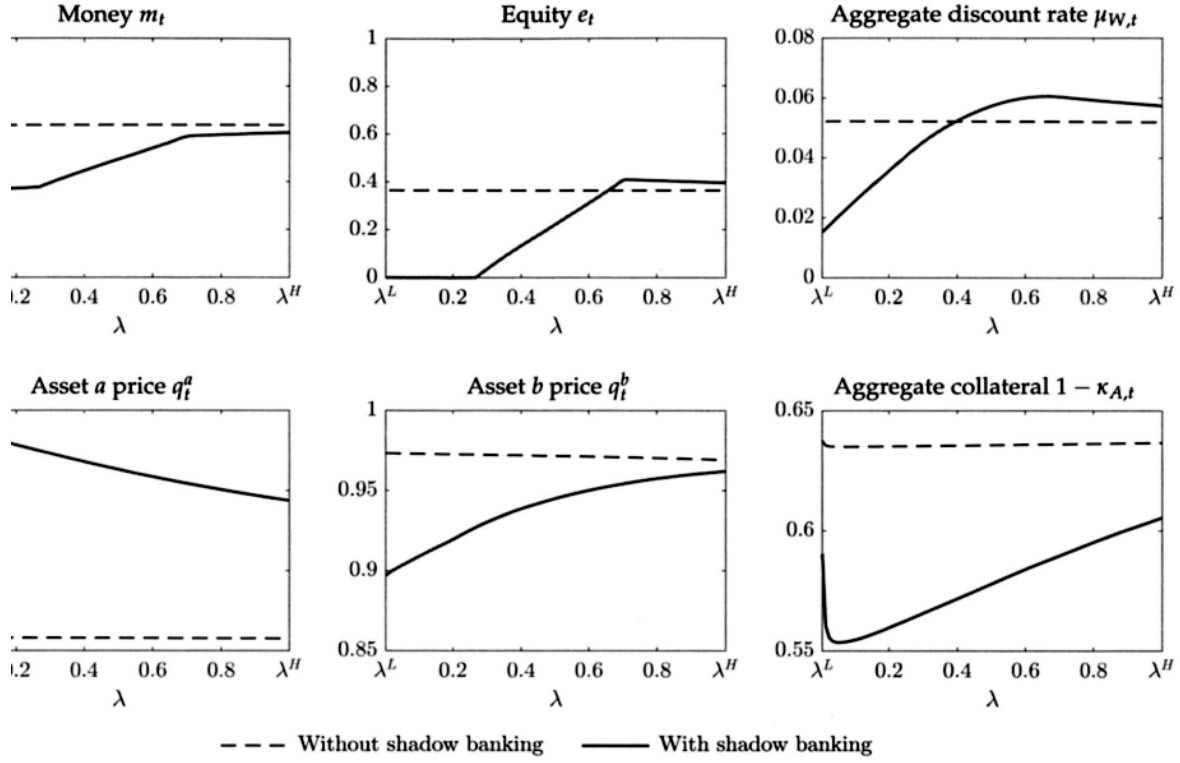


Figure 5: The economy with and without shadow banking.

6.6 Figure 6: Uncertainty Shock Response; Figure 7: Crash Response

Read: Uncertainty shocks widen shadow-money spreads and raise discount rates; crash shocks generate deeper and more persistent output losses.

Economic message: Fragile liquidity magnifies both news shocks and realized crashes.

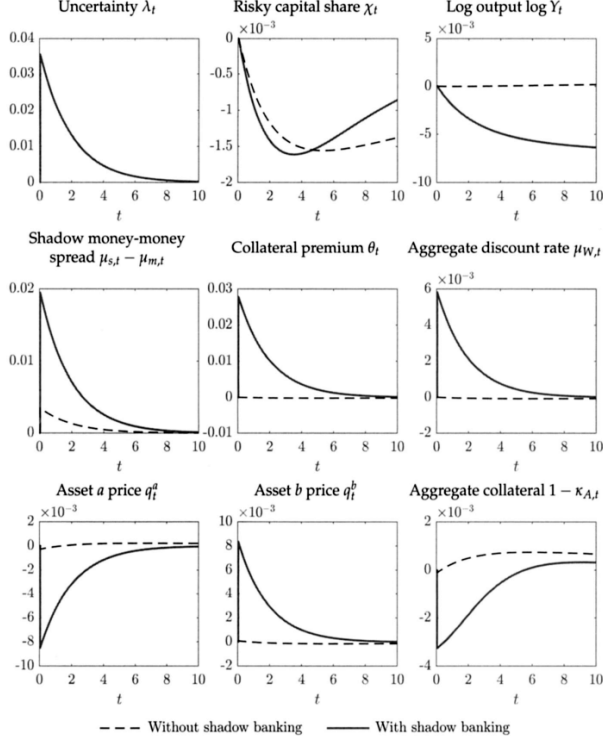


Figure 6. Uncertainty shock response. This figure shows impulse responses to an uncertainty shock to λ_t from its steady state of 0.0245 to 0.06 under the benchmark parameters in Table II. The initial risky capital share χ_t is in steady state. Responses are relative to trend. Solid red lines are for the economy with shadow banking. Dashed black lines are for the economy without shadow banking ($\bar{\kappa} = 0$). Log output is the cumulative sum of the expected growth rate of output. (Color figure can be viewed at wileyonlinelibrary.com)

(a) Figure 6: Uncertainty shock response.

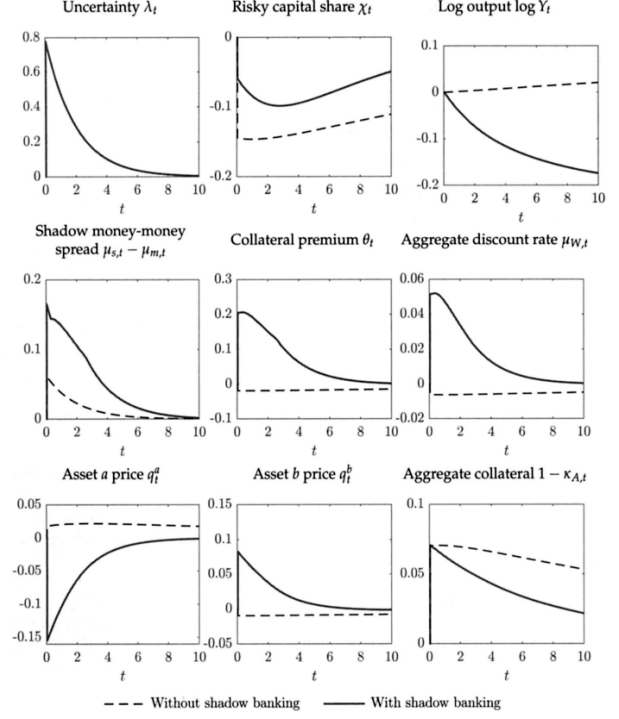


Figure 7. Crash response. This figure shows impulse responses to a crash shock when uncertainty λ_t and the risky capital share χ_t are in steady state. Responses are relative to trend. Solid red lines are for the economy with shadow banking, solved using the parameters in Table II. Dashed black lines are for the economy without shadow banking, solved using $\bar{\kappa} = 0$. Log output is the cumulative sum of the expected growth rate of output. (Color figure can be viewed at wileyonlinelibrary.com)

(b) Figure 7: Crash response.

6.7 Figure 8: Large Scale Asset Purchases (LSAP); Figure 9: Operation Twist

Read: LSAP supports risky asset prices by adding safe collateral after crashes, while Operation Twist can reduce useful collateral by replacing long safe assets with floating-rate safe assets.

Economic message: Central-bank balance-sheet policy should be judged by its effect on aggregate collateral, not only on safe yields.

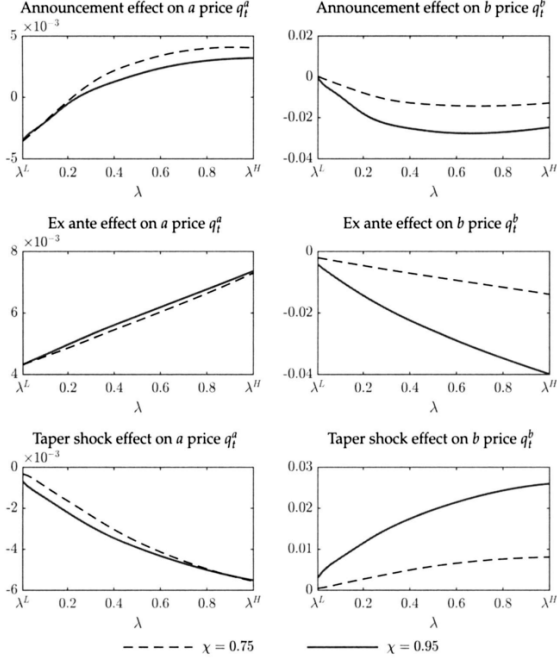


Figure 8. Large Scale Asset Purchases (LSAP). This figure shows the percentage change in asset prices around an LSAP intervention (announcement effect), between economies with and without the possibility of LSAP (ex ante effect), and around a taper shock. In an LSAP intervention, the central bank buys 20% of the stock of asset a and sells an equal dollar amount of asset b at postannouncement prices. LSAP interventions occur with probability one immediately after a crash. The announcement effect compares post-LSAP prices with prices when an LSAP fails to occur (it is still expected to occur at the next crash). The taper shock is an increase in the intensity of LSAP withdrawal from 0.1 to 1. See Section E.1 and Appendix E.1 for details. (Color figure can be viewed at wileyonlinelibrary.com)

(a) Figure 8: Large Scale Asset Purchases.

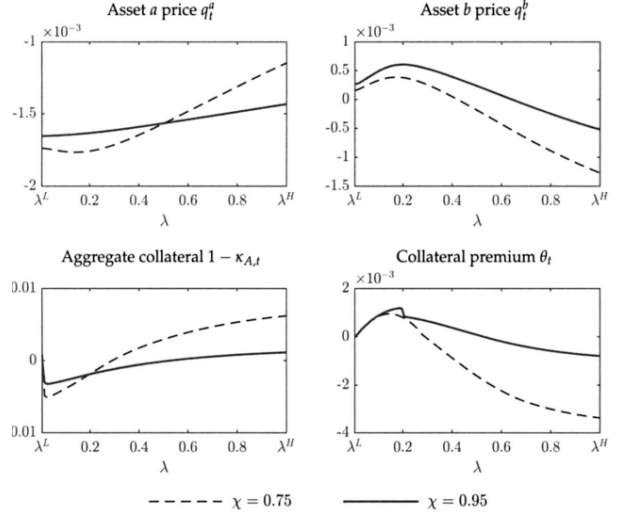


Figure 9. Operation Twist. This figure shows the percentage change in asset prices q_t^a and q_t^b and the change in aggregate collateral $1 - \kappa_{A,t}$ and the collateral premium θ_t around an Operation Twist intervention. In an Operation Twist intervention, the central bank purchases the economy of asset b funded with floating-rate (zero-duration) risk-free assets. The effects are computed by comparing quantities just before and just after an unanticipated Operation Twist intervention takes place. We set the rate of issuance of government bonds to 7%. See Section IV.B and Appendix E.1 for details. (Color figure can be viewed at wileyonlinelibrary.com)

(b) Figure 9: Operation Twist.

7 Short summary

- Shadow banking creates fragile liquidity by tranching risky assets into money-like claims.
- Low uncertainty supports shadow money, high asset prices, and growth.
- High uncertainty destroys shadow money, raises liquidity premia, and lowers investment.
- The model explains flight to quality, collateral runs, slow recoveries, LSAP, and Operation Twist.

8 Conclusion

8.1 Core conclusion

Shadow banking is a macroeconomic force because it expands liquidity in good times and contracts sharply when uncertainty rises.

8.2 Incremental conclusion

The paper's incremental contribution is to reinterpret crises as collapses in fragile liquidity rather than only intermediary net worth. This changes the implications for regulation and monetary policy.

8.3 Policy implications

Policies should be evaluated by whether they increase useful collateral and stable liquidity. LSAP can be expansionary; Operation Twist can be contractionary.

8.4 Limitations

The paper is theoretical and calibrated. It abstracts from many institutional details of shadow banking, regulation, heterogeneous intermediaries, and strategic runs.

8.5 Takeaway

The central lesson is that money-like claims can be both useful and dangerous. They are useful because they expand liquidity; they are dangerous because their liquidity can disappear precisely when the economy needs liquidity most.